

Shivam Kumar

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Summary

AI Engineer and Software Developer with over 2 years of experience building and deploying scalable AI-driven backend systems and machine learning applications. Skilled in designing and optimizing solutions across cloud platforms (AWS), with a proven record of developing high-impact applications like recommendation engines, retrieval-augmented generation (RAG), and LLM fine-tuning. Proficient in Python, SQL, and AWS, with hands-on expertise in creating automated workflows and enhancing data retrieval efficiency. Driven to apply advanced machine learning and data analysis skills to solve complex challenges and contribute to innovative, data-focused projects in a dynamic tech environment.

Professional Experience

YogLeads

Noida, India

Software Engineer

June 2023 - Oct 2024

- Engineered a Graph-RAG application using Neo4j, reducing query response time by 40% and streamlining data retrieval.
- Implemented NER in Neo4j to categorize 1,000+ nodes, enhancing data search and categorization.
- Created a RAG application with Llama Index and Qdrant DB, improving document retrieval efficiency by 30%.
- Deployed and fine-tuned Automatic Speech Recognition using Whisper and Nemo, achieving 95% accuracy in speech-to-text conversion for enhanced data processing.
- Spearheaded backend development, creating and deploying 100+ RESTful APIs, reducing data latency by 25% for web and mobile platforms using the FAST API framework.
- Built and managed WhatsApp automation using Selenium and AWS EC2 with a distributed, queue-based architecture, automating message dispatch and reducing manual intervention by 90%.
- Led team development efforts by conducting in-depth code reviews, improving code quality, and reducing bugs.

D Y Patil International University

Pune

Teaching Assistant Python

Aug 2022 - Jan 2023

- Facilitated learning for a class of 50+ first-year BTech CSE students, delivering structured Python programming lessons to enhance foundational understanding.
- Assessed and provided constructive feedback on 10+ assignments, driving a measurable improvement in coding accuracy and logical reasoning.
- Provided one-on-one mentoring and support, enhancing students' grasp of course material and problem-solving skills.
- Developed and delivered engaging lessons, achieving a 90% course completion rate and positive student feedback.
- Facilitated hands-on coding sessions, enabling students to successfully implement 15+ real-world Python projects.

Blackcoffer

Remote

Data Scientist Associate

April 2022 - June 2022

- Engaged with international clients to identify and resolve data challenges, facilitating 5+ project reviews and achieving a 90% client satisfaction score.
- Executed wine sales analysis, designing and implementing a predictive recommendation system that increased client revenue by 15% through targeted marketing.
- Designed and integrated a cross-platform Slack chatbot with Telegram, streamlining team communications and reducing response time by 20%.

Strata Unison, New York

Remote

Data Annotator Intern

Sep 2021 - Feb 2022

- Annotated over 5,000 images with 98% accuracy, delivering high-quality datasets that enhanced model training for image recognition projects.
- Employed Supervisely for efficient image annotation, reducing annotation time per image by 15% and ensuring consistent data quality for analysis.

Projects

Mood-based Music Recommendation System Using Brainwaves (Research Project)

- Crafted an EEG-based music recommendation system to enhance emotional well-being, leveraging real-time brainwave data from the Emotiv EPOC headset.
- Utilized Python for robust system development and integration. Applied statistical quantile methods for accurate mood classification from EEG datasets.
- Integrated Spotify dataset for personalized music recommendations aligned with user emotions. Used Selenium to automate recommended music playing on YouTube.
- The research paper can be accessed here: <https://ijrpr.com/uploads/V4ISSUE6/IJRPR14262.pdf>

Targeted Advertisement Using PySpark

- Devised a targeted advertisement system with PySpark, achieving a 20% improvement in processing efficiency by leveraging parallel data processing for faster, more accurate ad targeting.
- Formulated a recommendation engine that analyzes user watch time data, enabling personalized ad delivery and increasing click-through rates by 15%.
- Leveraged Python and PySpark for real-time data manipulation and analysis, supporting advanced personalization algorithms that matched user interests to ads effectively.
- Implemented collaborative filtering algorithms to align ads with user interests, resulting in a 12% engagement increase through tailored ad recommendations.

Covid19 Vaccination Management

- Developed a COVID-19 vaccination management system with PHP and MySQL, enabling efficient record-keeping and scheduling for over 1,500 university students.
- Refined an interface to streamline patient vaccination data entry, improving data accuracy and retrieval speed by 25% through efficient database structuring.
- Defined an intuitive, mobile-responsive interface for seamless vaccination record entry, reducing user input errors and improving data accuracy.
- Ensured secure storage and retrieval of patient data with MySQL, implementing data encryption and access controls to comply with healthcare data standards.
- Facilitated real-time vaccine status tracking and scheduling, reducing administrative processing time by 20% and enhancing patient compliance tracking.

Education

D Y Patil International University

Aug 2019 - June 2023

B.Tech in Computer Science and Engineering

- Percentage: 8 CGPA
- Specialization: Data Science and Machine Learning

Skills

Programming Languages: Python, Java

Database Technologies: MySQL, Redis, DynamoDB, MongoDB

Frameworks & Libraries: Flask, FastAPI, Django, TensorFlow, PyTorch, Langchain, Llama Index

Cloud Platforms & Services: AWS (EC2, SQS, CloudWatch, SES, OpenSearch, S3, Lambda)

DevOps & Monitoring Tools: Git, Docker, New Relic, PM2

Data Science & Analytics: Power BI, Tableau, Data Analysis, Data Engineering, Machine Learning, Neural Networks

Soft Skills: Analytical Thinking, Leadership, Communication, Collaboration

Languages: English, Hindi, Japanese

Achievements

Data Science Code-a-thon Contest (TECHGIG)

Ranked 1st among competitors in TechGig's Data Science Code-a-thon, recognized for outstanding contributions and innovative solutions in data science.

Gold Badge – Python, SQL, and Problem Solving (HackerRank)

Awarded 5-Star Gold Badge in Python and SQL, and 6-Star Gold Badge in Problem-Solving on HackerRank, demonstrating advanced skills in complex query-solving, analytical reasoning, and programming proficiency.

LeetCode Performance

Successfully solved 600+ problems, showcasing consistency in daily learning and a strong commitment to honing algorithmic and problem-solving skills.

Certifications

Microsoft AI Classroom Series - Microsoft 

Design Thinking and Global Startup - Korea Advanced Institute of Science and Technology 

Design Thinking for Innovation - University of Virginia 

Socials

Github: <https://github.com/data-scientist-shivam799>

LinkedIn: <https://www.linkedin.com/in/entrepreneur-shivam/>

HackerRank: <https://www.hackerrank.com/profile/shivam799kumar>